AZAZ UR REHMAN

Full Stack Developer

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PROFESSIONAL SUMMARY

Computer Science graduate from NUTECH with hands-on experience across the full stack — backend development in Python (FastAPI, Flask) and Node.js (Express.js), and frontend development using React.js. Skilled in designing and consuming REST APIs, managing PostgreSQL/MySQL databases, and shipping complete, deployed applications end-to-end. Comfortable working across the stack and quick to pick up new frameworks. Additional expertise in AI, Machine Learning, and Computer Vision. Passionate about writing clean, maintainable code and continuously learning modern web technologies.

EDUCATION

Bachelor of Science in Computer Science
National University of Technology (NUTECH) | Nov 2021 – Jun 2025 | Islamabad | GPA: 3.05

PROFESSIONAL EXPERIENCE

Associate Software Developer | *NUTECH Innovation Center (NIC)* | *Mar 2026 – Sep 2026* | *Islamabad*

- Integrated offline maps into an Android application, plotting points and diagrams on the map from user-provided coordinates (Eastings and Northings).
- Worked with Heltec modules, including module configuration and offline data communication between host and station devices, and built the companion Android app for this communication.
- Created an AI-based weather prediction system by training a model on weather data to forecast temperature, pressure, wind speed, and air density.

Associate AI Engineer | *Cware Technologies* | *Nov 2025 – May 2026* | *Islamabad (remote)*

- Developed scalable backend applications and REST APIs using Python, FastAPI, and PostgreSQL.
- Built a full-stack Todo application using Node.js and Express.js for the REST API backend and React.js for the frontend, with endpoints tested and documented via Swagger.
- Built secure authentication and authorization systems using JWT.
- Designed and optimized PostgreSQL databases with Alembic migrations.
- Implemented backend optimization, API testing, and database migrations.
- Successfully delivered projects: SIMAPP, CardTopup, and GitHub Automation.

Associate Deep Learning Engineer | *IT Solera* | *Jun 2025 – Aug 2025* | *Islamabad (remote)*

- Built deep learning models using TensorFlow and PyTorch for computer vision applications.
- Performed image preprocessing, data augmentation, model training, and evaluation.

AI & ML Intern | *National Disaster Management Authority (NDMA)* | *Jul 2024 – Aug 2024* | *Islamabad*

- Assisted in AI/ML research, data analysis, and reporting.
- Developed analytical reports using real-world disaster management data.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, Java, C#

Frontend Development: React.js, Vite

Backend Development: Node.js, Express.js, FastAPI, Flask, REST API Development, JWT Authentication

Databases: PostgreSQL, MySQL, SQLite, Alembic, JSONB, Prisma

API Testing & Docs: Swagger

AI / Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, YOLO, OpenCV, DenseNet, ResNet, EfficientNet

Version Control: Git, GitHub

PROJECTS

AI-Based Weather Prediction System | *Completed*
Built a weather prediction system: downloaded and processed ERA5 reanalysis weather data from an online source using Python, and trained a SimVP (Simple Video Prediction) model from scratch on the dataset to forecast short-term weather conditions.
Technologies: Python, ERA5 Dataset, SimVP, Deep Learning

WordPress Product Image Automation Tool
Built a local Windows tool that automates product image listing for a WooCommerce store. It finds a product, reads the original image source from its description, collects candidate images with a headless browser, uses Claude (Anthropic API) to pick the best 4–5, processes them, and uploads them as the main image and gallery. Includes a human approval step before publishing, auto-publish off by default, and a batch mode that processes matching draft products one by one.
Technologies: Python, FastAPI, SQLAlchemy, SQLite, Playwright, OpenCV, Pillow, Anthropic API, React, Vite, TypeScript, Tailwind CSS

Aurelia Jewels – MERN Stack E-Commerce Store
Built a full-stack online jewellery store with the MERN stack. Customers can sign up, log in, browse products, add to cart, and place orders, with email notifications sent to both the customer and the admin. A single-admin panel lets the store owner add, edit, and delete products, track stock, and view recent customer orders. Login and signup use an animated, interactive UI built with GSAP.
Technologies: MongoDB, Express.js, React.js, Node.js, GSAP

Waypoint – Fuel Route Planning App | *Deployed*
Built and shipped a fully deployed full-stack application: users enter a starting point and destination, and the app plots the optimal driving route while finding the cheapest fuel stops genuinely along the path, ranked by price per gallon. Designed the REST API for geocoding, route computation, and fuel-stop optimization, and built the React frontend with live interactive routing and mapping.
Technologies: Django REST Framework, React, Vite, Mapbox GL, Render, Vercel

Todo Application – Full Stack
Built a complete Todo application with a REST API backend using Node.js and Express.js, and a React.js frontend. Implemented full CRUD functionality connecting frontend to backend, and tested and documented all endpoints using Swagger.
Technologies: Node.js, Express.js, React.js, REST API, Swagger

SIMAPP – Multi-Industry Management Platform | *Nov 2025 – May 2026*
Designed and developed the complete backend for a multi-industry management platform supporting Telecommunication, Event Management, and Logistics. Implemented secure user authentication, advanced search, QR/Barcode generation, and industry-specific workflows.
Technologies: Python, FastAPI, PostgreSQL, JWT, Alembic, JSONB, REST APIs

Web Scraper for Crop Prices (AMIS Punjab) | *Deployed*
Built a web scraping pipeline to collect daily wholesale crop prices from AMIS Punjab across 80+ mandis and 5 major crops, converting the data into a clean, bilingual (English/Urdu) API for consumption by a mobile app. Implemented data cleaning, deduplication, and structured storage to handle inconsistent market names and missing or duplicate records.
Technologies: Python, Web Scraping, REST API, Data Cleaning

AI Resume Builder | *Jan 2026*
Developed an AI-powered resume builder that generates professional resumes using LLMs. Integrated FAISS for semantic search and intelligent content recommendations. Built backend APIs for resume generation, storage, and user interaction.
Technologies: Python, OpenAI API, FAISS, FastAPI, PostgreSQL

Vehicle Detection & Counting System | *Mar 2026*
Developed a real-time vehicle detection and counting system using YOLO and OpenCV for traffic monitoring. Implemented object detection and tracking to count vehicles from video streams.
Technologies: Python, YOLO, OpenCV

CERTIFICATIONS

AI and Data Science Certification – NAVTTC, Pakistan